

Conditional Fiber Geometry for Incomplete Inverse Data

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Abstract

Let $V \in E \subset \mathbb{R}^d$ parameterize a finite-dimensional inverse problem, let $X = F(V)$ denote its complete data, and let a mask M reveal $P_M X$. We study the conditional geometry induced by this incomplete observation. Near a compact regular observation fiber with Jacobian rank r , the ambient level set has dimension $d - r$, and the support of the associated conditional data law has covering upper exponent at most $d - r$. For noisy observations, we derive local covering estimates that retain every nonzero singular value of the masked, noise-whitened Jacobian. On a compact regular region satisfying boundary transversality and a bidirectional no-exit condition, a pseudoinverse horizontal flow transports complete fibers with displacement bounded by $\|z - z'\|/s_0$. Coarea disintegration then yields quantitative total-variation control after transport and Wasserstein-1 continuity of the conditional data laws. We further establish motion and curvature estimates for noncollapsed image manifolds and L^1 regularity after Gaussian smoothing. Experiments on a discretized quadrilateral EIT model examine these mechanisms independently. Two fiber-construction procedures recover PCA dimensions 6, 4, and 2 for numerical ranks 2, 4, and 6; a separate rank-eight experiment exhibits an isolated local solution. Additional tests support the predicted δ/σ_i scaling and the use of the full singular spectrum. The numerical results do not establish differential rank, global flow completeness for the polygon class, continuum finite-element convergence, conditional-score regularity, or a uniform diffusion-model rate.

1 Introduction

Incomplete measurements constrain an inverse-problem parameter without, in general, identifying it. The remaining feasible parameters form an observation fiber, and its image under the complete-data map contains the support of the corresponding conditional data law. This elementary description becomes useful when the complexity of a conditional generative model is controlled through metric entropy. It also separates two properties that measurement count alone cannot distinguish: residual tangent dimension and stability in the observed normal directions.

This work is motivated by polygonal electrical impedance tomography (EIT), where data completion has been studied both as a computational device [4] and through conditional diffusion models [5]. Finite-measurement identifiability and stability depend on the admissible conductivity class and measurement configuration [3, 6]. Jacobian singular values, Fisher information, sensitivity-volume criteria, and Bayesian experimental design are established tools in EIT [7, 8, 11, 12]. Conditional distributions on regular level sets are likewise classical consequences of disintegration and the coarea formula and have recently been used explicitly in conditional generative modeling [1]. These ingredients are used here as background, not presented as new constructions.

The EIT analysis of Chen et al. [5] controls the conditional law by the entropy of the global polygonal Dirichlet-to-Neumann (DtN) class. Its conditional diffusion bound therefore retains

dimension $2n_v$ for every observation of an n_v -vertex polygon. We replace that global set by

$$F(E \cap (P_m F)^{-1}(z)),$$

the image of the actual observation fiber. The resulting analysis connects masked inverse observations, fiber entropy, and the intrinsic covering input used by an external diffusion convergence theorem.

The contributions are as follows.

1. We derive compact regular-fiber covering bounds and formulate the corresponding conditional-support result for a random family of masks.
2. We establish linear and local nonlinear noisy-tube bounds that retain the complete nonzero singular spectrum of the observation Jacobian, including its noise-whitened form.
3. Under explicit boundary and flow-completeness assumptions, we construct a quantitative pseudoinverse transport between fibers and prove stability of the coarea conditional measures and their complete-data pushforwards.
4. We apply the entropy bound pointwise to the version-pinned theorem of Liang et al. [10, arXiv version 2]. This is a conditional corollary of an external result, not an observation-uniform convergence theorem.
5. We provide numerical EIT experiments designed to test the local geometric mechanisms and their failure modes in a finite-dimensional discretization.

All theoretical laws below are population laws. A finite empirical training set has finite support and hence zero covering exponent below its minimum sample separation; it is not the object of the continuum entropy theorem. The abstract results hold under the stated hypotheses. Their global validity over the complete polygon prior used in EIT is not asserted.

Section 2 introduces the mathematical setting. Sections 3 and 4 treat exact fibers and noisy tubes. Sections 5 and 6 develop fiber and measure transport, Section 7 considers image manifolds and Gaussian smoothing, and Section 8 states the external diffusion consequence. Sections 9 and 10 describe the EIT model, numerical evidence, and supporting interface checks.

2 Mathematical setting

Let $O \subset \mathbb{R}^d$ be open, $E \Subset O$ compact, and $F \in C^1(O; \mathbb{R}^D)$. A finite or countable mask family $\mathcal{M}$ contains linear maps $P_m : \mathbb{R}^D \rightarrow \mathbb{R}^{q_m}$. Put

$$H_m = P_m F, \quad S_{m,z} = E \cap H_m^{-1}(z).$$

Equip the countable disjoint union

$$Z = \bigsqcup_{m \in \mathcal{M}} (\{m\} \times \mathbb{R}^{q_m})$$

with its standard Borel structure. Let (V, M) be any jointly distributed $E \times \mathcal{M}$ -valued random element; independence is unnecessary. Define

$$X = F(V), \quad Y = P_M X, \quad Z = (M, Y) \in Z.$$

Covering numbers $\mathcal{N}_\varepsilon(A)$ use Euclidean balls whose centers may lie in the ambient space; requiring centers in A changes only fixed radius constants. We use $\text{TV}(\mu, \nu) = \sup_B |\mu(B) - \nu(B)|$, equivalently one half of the L^1 distance when densities exist. All singular values refer to explicitly fixed Euclidean parameter and observation norms. Under noise covariance Γ , the observation norm is whitened and the relevant Jacobian is $\Gamma^{-1/2} D H$.

3 Conditional support and exact-fiber entropy

Theorem 3.1 (Compact regular-fiber entropy). *Fix a linear $P : \mathbb{R}^D \rightarrow \mathbb{R}^q$, put $H = PF$, and let $S_z = E \cap H^{-1}(z)$ be nonempty. Suppose an open U satisfies $S_z \subset U \subset O$ and $\text{rank } DH(v) = r$ for every $v \in U$. Set $k = d - r$. Then $H^{-1}(z) \cap U$ is an embedded C^1 submanifold of dimension k . If $k \geq 1$, there are $C_z < \infty$ and $\varepsilon_z > 0$ such that*

$$\mathcal{N}_\varepsilon(F(S_z)) \leq C_z \varepsilon^{-k}, \quad 0 < \varepsilon \leq \varepsilon_z. \quad (1)$$

If $k = 0$, S_z and $F(S_z)$ are finite.

Proof. The constant-rank level-set theorem makes $Q_z = H^{-1}(z) \cap U$ an embedded C^1 manifold of dimension k [9]. Every $v \in S_z$ has a relatively compact chart whose inverse parametrization $\psi_v : B_k(0, a_v) \rightarrow Q_z$ has bounded derivative on a closed smaller ball. The composition $F \circ \psi_v$ is C^1 there; its derivative is bounded on that closed Euclidean ball, which is convex, so the composition is Lipschitz. Compactness of S_z selects finitely many charts, indexed by $j = 1, \dots, L$. If K_j is the resulting Lipschitz constant, the volumetric Euclidean bound gives

$$\mathcal{N}_\varepsilon(F(S_z \cap \psi_j(B_k(0, a_j)))) \leq \left(1 + \frac{2a_j K_j}{\varepsilon}\right)^k.$$

Summing the finite covers proves (1) after restricting ε . For $k = 0$, Q_z is discrete, and its compact subset S_z is finite. $\square$

Proposition 3.2 (Random-mask conditional support). *Let $\nu_{m,z}$ be a regular conditional distribution of V given $Z = (m, z)$. For $\mathcal{L}(Z)$ -almost every (m, z) ,*

$$\nu_{m,z}(S_{m,z}) = 1, \quad \text{supp } \mathcal{L}(X \mid M = m, Y = z) \subseteq F(S_{m,z}).$$

Proof. The parameter, mask, and observation spaces are standard Borel, so let $\kappa_{m,z}$ be a regular conditional distribution of (V, M) given Z . For the measurable map $T(v, m) = (m, P_m F(v))$, the identity $Z = T(V, M)$ implies $T_\# \kappa_{m,z} = \delta_{(m,z)}$ for $\mathcal{L}(Z)$ -almost every (m, z) . This follows first on a countable generator of the disjoint-union Borel sigma-field and then by the monotone-class theorem. Consequently $\kappa_{m,z}$ is concentrated on $\{(v, m) : H_m(v) = z\}$. Its V -marginal is $\nu_{m,z}$, so $\nu_{m,z}(S_{m,z}) = 1$. The conditional law of X is $F_\# \nu_{m,z}$; since $S_{m,z}$ is compact, $F(S_{m,z})$ is closed, proving the support inclusion. $\square$

Corollary 3.3 (Random-mask conditional entropy). *Suppose $d = 2n_v$ and, for $\mathcal{L}(Z)$ -almost every relevant (m, z) , DH_m has constant rank $r(m, z)$ on a neighborhood of the nonempty compact fiber. When $r(m, z) < 2n_v$, a constant $C''_{m,z} < \infty$ satisfies*

$$\log \mathcal{N}_\varepsilon(\text{supp } \mathcal{L}(X \mid M = m, Y = z)) \leq C''_{m,z} + [2n_v - r(m, z)] \log(1/\varepsilon), \quad 0 < \varepsilon \leq 1.$$

If $r(m, z) = 2n_v$, the conditional support is finite.

Proof. Combine Proposition 3.2 and Theorem 3.1. Put $\varepsilon_* = \min\{1, \varepsilon_{m,z}\}$. The theorem gives the stated form below ε_* after taking logarithms. On $[\varepsilon_*, 1]$, monotonicity gives $\log \mathcal{N}_\varepsilon \leq \log \mathcal{N}_{\varepsilon_*} < \infty$; enlarge the additive constant to cover this compact scale interval. $\square$

This is an upper bound. It need not be sharp because F can collapse fiber directions and the conditional probability may occupy only part of the image.

4 Quantitative local geometry of noisy fibers

Theorem 4.1 (Linear full-spectrum tube). *Let $J \in \mathbb{R}^{q \times d}$ have rank $r \geq 1$ and nonzero singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$. For*

$$\mathcal{T}_{z,\delta}(R) = \{x \in B_d(0, R) : \|Jx - z\|_2 \leq \delta\},$$

if the set is nonempty, a constant C_d depending only on d satisfies

$$\mathcal{N}_\varepsilon(\mathcal{T}_{z,\delta}(R)) \leq \left(1 + \frac{C_d R}{\varepsilon}\right)^{d-r} \prod_{i=1}^r \left(1 + \frac{C_d \delta}{\sigma_i \varepsilon}\right), \quad \varepsilon > 0. \quad (2)$$

Proof. Take $J = U_r \Sigma_r V_r^\top$ and complete V_r by V_0 . Choose $x_0 \in \mathcal{T}_{z,\delta}(R)$. Write $x = V_r a + V_0 b$ and $x_0 = V_r a_0 + V_0 b_0$. Then $\|b\| \leq R$ and the two residual bounds imply

$$\|\Sigma_r(a - a_0)\|_2 = \|J(x - x_0)\|_2 \leq 2\delta.$$

Thus the tube lies in the product of a $(d - r)$ -ball of radius R and an ellipsoid with semiaxes $2\delta/\sigma_i$. Cover the two factors at radius $\varepsilon/\sqrt{2}$ and take products. Standard ball and axis-ellipsoid packing bounds [15, Chapter 4] give (2), after changing C_d . If centers must lie in the tube, use a maximal separated subset of the tube and bound it by the same containing-product packing number. $\square$

Corollary 4.2 (Noise-whitened tube). *For symmetric positive-definite Γ , replace the residual by $\|\Gamma^{-1/2}(Jx - z)\| \leq \delta$. Then Theorem 4.1 holds with the singular values of $\Gamma^{-1/2}J$.*

Proof. Apply Theorem 4.1 to $\tilde{J} = \Gamma^{-1/2}J$ and $\tilde{z} = \Gamma^{-1/2}z$. $\square$

When $\delta > 0$ and $\varepsilon \ll \delta/\sigma_i$ for every $i = 1, \dots, r$, the normal logarithmic contribution contains

$$r \log(\delta/\varepsilon) - \sum_{i=1}^r \log \sigma_i + O_d(1).$$

For full row rank, $\sum_i \log \sigma_i = \frac{1}{2} \log \det(JJ^\top)$. This does not introduce D-optimality; it gives the classical EIT sensitivity-volume criterion [12] a conditional-covering interpretation.

Lemma 4.3 (Quantitative implicit graph in singular coordinates). *Let $k = d - r$, let G be C^2 on a neighborhood of $\overline{B}_d(0, 2R)$, and suppose*

$$G(0, 0) = 0, \quad D_a G(0, 0) = \Sigma = \text{diag}(\sigma_1, \dots, \sigma_r), \quad D_b G(0, 0) = 0,$$

where $\sigma_1 \geq \dots \geq \sigma_r = s > 0$. Suppose also that

$$\|DG(x) - DG(y)\|_{\text{op}} \leq L\|x - y\|, \quad x, y \in \overline{B}_d(0, 2R), \quad LR \leq s/16.$$

Then there is a unique continuous map $g : \overline{B}_k(0, R) \rightarrow B_r(0, R/8)$ satisfying $G(g(b), b) = 0$; its restriction to $B_k(0, R)$ is C^1 . It obeys

$$\|g(b)\| \leq \frac{L}{s}\|b\|^2, \quad \|Dg(b)\|_{\text{op}} \leq \frac{17}{239} < \frac{1}{8}. \quad (3)$$

If $\|(a, b)\| \leq R$, $\|G(a, b)\| \leq \delta$, and $c = a - g(b)$, then

$$\|c\| \leq \frac{8\delta}{7s}, \quad \|\Sigma c\| \leq \frac{8\delta}{7}. \quad (4)$$

Proof. For fixed $\|b\| \leq R$, define $\mathcal{T}_b(a) = a - \Sigma^{-1}G(a, b)$ on $\overline{B}_r(0, R/8)$. If $\|a\| \leq R/8$, then $\|(a, b)\| < 17R/16$, and hence

$$\|D_a \mathcal{T}_b(a)\|_{\text{op}} \leq \frac{L}{s} \|(a, b)\| \leq \frac{17}{256}.$$

Taylor expansion along $t \mapsto (0, tb)$, using $D_b G(0, 0) = 0$, gives

$$\|G(0, b)\| \leq \frac{L}{2} \|b\|^2, \quad \|\mathcal{T}_b(0)\| \leq R/32.$$

Consequently

$$\|\mathcal{T}_b(a)\| \leq R/32 + (17/256)(R/8) < R/8.$$

The contraction theorem gives a unique fixed point $g(b)$ in this ball, and

$$\|g(b)\| \leq \frac{1}{1 - 17/256} \frac{L\|b\|^2}{2s} \leq \frac{L}{s} \|b\|^2 \leq R/16.$$

The parameter-dependent contraction theorem makes g continuously differentiable. At $(g(b), b)$, both $\|D_a G - \Sigma\|$ and $\|D_b G\|$ are at most $17LR/16$. Implicit differentiation therefore yields

$$\|Dg(b)\|_{\text{op}} \leq \frac{17LR/16}{s - 17LR/16} \leq \frac{17}{239}.$$

For the final assertion, the graph point $(g(b), b)$ lies in $B_d(0, 17R/16)$. The segment from it to (a, b) remains in that convex ball. The fundamental theorem of calculus gives

$$G(a, b) - G(g(b), b) = (\Sigma + E)c, \quad E = \int_0^1 [D_a G(g(b) + tc, b) - \Sigma] dt, \quad \|E\| \leq s/8.$$

Since $G(g(b), b) = 0$, the smallest singular value bound gives $\|c\| \leq 8\delta/(7s)$. Substitution back gives

$$\|\Sigma c\| \leq \|(\Sigma + E)c\| + \|E\|\|c\| \leq \delta + \delta/7 = 8\delta/7,$$

which proves (4). $\square$

Theorem 4.4 (Quantitative local nonlinear spectral tube). *Let $R > 0$. Suppose H is C^2 and F is C^1 on a neighborhood of $\overline{B}(v_0, 2R)$, with*

$$\sup_{\overline{B}(v_0, 2R)} \|D^2 H\|_{\text{op}} \leq L_H, \quad \sup_{\overline{B}(v_0, 2R)} \|DF\|_{\text{op}} \leq L_F.$$

Suppose $J_0 = DH(v_0)$ has rank at least $r \geq 1$, with leading singular values $\sigma_1 \geq \dots \geq \sigma_r = s > 0$, and assume

$$L_H R \leq s/16, \quad 0 \leq \delta \leq sR.$$

Then, for

$$Q_{v_0, \delta} = \overline{B}(v_0, R) \cap \{v : \|H(v) - H(v_0)\| \leq \delta\},$$

a constant C_d depending only on d satisfies, for every $\varepsilon > 0$,

$$\mathcal{N}_\varepsilon(F(Q_{v_0, \delta})) \leq \left(1 + \frac{C_d L_F R}{\varepsilon}\right)^{d-r} \prod_{i=1}^r \left(1 + \frac{C_d L_F \delta}{\sigma_i \varepsilon}\right). \quad (5)$$

No constant-rank assumption is required for this upper bound.

Proof. Take the leading singular vectors in a thin SVD $J_0^\top U_r = V_r \Sigma_r$, complete V_r orthogonally by V_0 , and define

$$G(a, b) = U_r^\top [H(v_0 + V_r a + V_0 b) - H(v_0)].$$

Then $D_a G(0, 0) = \Sigma_r$, $D_b G(0, 0) = 0$, and the orthogonal projection and coordinate changes preserve the derivative Lipschitz bound L_H . Apply Lemma 4.3. For every point of $Q_{v_0, \delta}$, $b \in \overline{B}_{d-r}(0, R)$ and, with $c = a - g(b)$,

$$c \in E_\delta := \{c : \|\Sigma_r c\| \leq 8\delta/7\}.$$

Thus $Q_{v_0, \delta}$ is contained in the image of

$$\Theta(b, c) = v_0 + V_r(g(b) + c) + V_0 b, \quad (b, c) \in \overline{B}_{d-r}(0, R) \times E_\delta.$$

The assumption $\delta \leq sR$ and (3)–(4) imply that this entire parametrized containing set lies in $B(v_0, 2R)$. Moreover, $\text{Lip}(\Theta) \leq 2$. The ball is convex, so the displayed derivative bound makes $F \circ \Theta$ $2L_F$ -Lipschitz there.

Cover $B_{d-r}(0, R)$ and the axis ellipsoid E_δ separately in the product Euclidean metric. Their standard volumetric covering bounds are a tangent factor of order $(1 + C_d R/\eta)^{d-r}$ and normal factors $\prod_i (1 + C_d \delta/(\sigma_i \eta))$. Take $\eta = \varepsilon/(2L_F)$; if $L_F = 0$, the image is a singleton and the claim is immediate. Mapping the product cover through $F \circ \Theta$ proves (5). If covering centers are required to belong to $F(Q_{v_0, \delta})$, select a point from each intersecting ambient cover ball and absorb the resulting factor two into C_d . $\square$

Corollary 4.5 (Uniform exact-fiber cover on a regular region). *Fix $1 \leq r \leq \min\{d, q\}$. Let $K \subset B_d(0, D)$ be compact and let $\rho_0 > 0$. Let U_0 be an open neighborhood of the closed ρ_0 -neighborhood of K . Suppose $H \in C^2(U_0; \mathbb{R}^q)$ and $F \in C^1(U_0; \mathbb{R}^D)$ satisfy uniform bounds $\|D^2 H\| \leq L_H$ and $\|DF\| \leq L_F$ there, and $\sigma_r(DH(v)) \geq s > 0$ uniformly on K . Put*

$$R = \min\{\rho_0/2, s/(16L_H)\},$$

with $s/(16L_H) = +\infty$ when $L_H = 0$. Then

$$\mathcal{N}_\varepsilon(F(K \cap H^{-1}(z))) \leq \left(1 + \frac{C_d D}{R}\right)^d \left(1 + \frac{C_d L_F R}{\varepsilon}\right)^{d-r}$$

for every $z \in \mathbb{R}^q$ and $\varepsilon > 0$.

Proof. Choose a maximal R -separated subset of the compact fiber. Its cardinality is at most $(1 + C_d D/R)^d$ by a volumetric packing bound. The associated closed R -balls cover the fiber. Apply Theorem 4.4 with $\delta = 0$ at every center and sum the local covers. $\square$

5 Quantitative transport of fibers

Definition 5.1 (Bidirectional no-exit property). Let H have full row rank on a neighborhood of $\mathcal{K}_G = \mathcal{K} \cap H^{-1}(G)$. The pair $(\mathcal{K}, G)$ has the bidirectional no-exit property for the minimum-norm horizontal lift if, for every ordered pair $z, z' \in G$ and every $v_0 \in S_z$, the initial-value problem

$$\dot{v}(t) = DH(v(t))^\dagger(z' - z), \quad v(0) = v_0,$$

has a unique solution for every $t \in [0, 1]$ and that solution remains in $\mathcal{K}$. Because the definition quantifies over ordered pairs, it also applies with z and z' exchanged. Convexity of G then ensures $H(v(t)) = (1 - t)z + tz' \in G$ along every such solution.

Assumption 5.2 (Quantitative regular observation tube). *Let $\mathcal{U} \subset \mathbb{R}^d$ be open, $\mathcal{K} \Subset \mathcal{U}$ a compact d -dimensional C^2 submanifold with boundary, and let $G \subset \mathbb{R}^r$ be nonempty, compact, and convex. Put $\mathcal{K}_G = \mathcal{K} \cap H^{-1}(G)$ and fix one open neighborhood $\mathcal{U}_Q$ of $\mathcal{K}_G$ with $\overline{\mathcal{U}_Q} \Subset \mathcal{U}$. All suprema and infima below are taken over this same fixed set $\mathcal{U}_Q$. Assume:*

1. $H \in C^2(\mathcal{U}; \mathbb{R}^r)$ has full row rank on $\mathcal{U}_Q$, $s_0 = \inf_{\mathcal{U}_Q} \sigma_r(DH) > 0$, and $L_H = \sup_{\mathcal{U}_Q} \|D^2H\|_{\text{op}} < \infty$;
2. the boundary restriction is also a submersion:
$$DH(v)(T_v \partial \mathcal{K}) = \mathbb{R}^r, \quad v \in \partial \mathcal{K} \cap H^{-1}(G).$$
3. $F \in C^2(\mathcal{U}; \mathbb{R}^D)$, with $M_F = \sup_{\mathcal{U}_Q} \|DF\|_{\text{op}}$ and $M_{F,2} = \sup_{\mathcal{U}_Q} \|D^2F\|_{\text{op}}$ finite;
4. V has a Lebesgue density ρ on $\mathcal{K}$ whose restriction to the regular tube extends to a positive C^1 function on $\mathcal{U}_Q$, with $0 < \rho_{\min} = \inf_{\mathcal{U}_Q} \rho \leq \sup_{\mathcal{U}_Q} \rho = \rho_{\max} < \infty$ and $L_\rho = \sup_{\mathcal{U}_Q} \|\nabla \rho\| < \infty$;
5. every $S_z = \mathcal{K} \cap H^{-1}(z)$, $z \in G$, is nonempty and has positive $(d-r)$ -dimensional Hausdorff measure;
6. $(\mathcal{K}, G)$ has the bidirectional no-exit property of Definition 5.1.

Write $k = d - r$, $D_{\mathcal{K}} = \text{diam}(\mathcal{K})$, and $D_F = \text{diam } F(\mathcal{K}_G)$.

The boundary-submersion hypothesis and the regular-level-set theorem with boundary make every S_z a neat C^2 submanifold of $\mathcal{K}$, with $\partial S_z = S_z \cap \partial \mathcal{K}$. It excludes boundary tangencies that can add isolated points to an otherwise positive-dimensional fiber. The no-exit property is a separate, essential flow-completeness hypothesis. The stronger condition

$$\text{range } DH(v)^\dagger \subseteq T_v \partial \mathcal{K}, \quad v \in \partial \mathcal{K} \cap H^{-1}(G),$$

is sufficient both for boundary submersion and for invariance of the boundary under every horizontal field, but it is not imposed when no-exit is verified by another argument.

Lemma 5.3 (Derivative of the full-row-rank right inverse). *For full-row-rank $B \in \mathbb{R}^{r \times d}$, let $A = B^\dagger = B^\top (BB^\top)^{-1}$. For perturbation E ,*

$$DA[B][E] = -AEA + (I_d - AB)E^\top A^\top A.$$

Under Assumption 5.2,

$$\|DA(v)[h]\|_{\text{op}} \leq C_A \|h\|, \quad C_A = \frac{2L_H}{s_0^2}.$$

Proof. Differentiate $B^\top (BB^\top)^{-1}$, use $D(C^{-1})[K] = -C^{-1}KC^{-1}$, and regroup using $BA = I_r$ and symmetry of AB . Because AB is an orthogonal projector and $\|A\| = 1/\sigma_r(B)$, each of the two terms is bounded by $\|E\|/\sigma_r(B)^2$. Substitute $E = D^2H(v)[h]$. $\square$

Theorem 5.4 (Quantitative fiber transport). *Under Assumption 5.2, for $z, z' \in G$ put $\xi = z' - z$ and solve*

$$\dot{v}(t) = DH(v(t))^\dagger \xi, \quad v(0) = v.$$

Each S_z is a compact neat C^2 manifold with boundary. The time-one map $\Phi_{z,z'} : S_z \rightarrow S_{z'}$ is a C^1 diffeomorphism with inverse $\Phi_{z',z}$. Writing $\Delta = \|z' - z\|$,

$$\|\Phi_{z,z'} - \text{Id}\|_\infty \leq \Delta/s_0, \quad (6)$$

$$\|D\Phi_{z,z'}\|_{\text{op}} \leq e^{C_A\Delta}, \quad (7)$$

$$\|D\Phi_{z,z'} - I_d\|_{\text{op}} \leq e^{C_A\Delta} - 1, \quad (8)$$

$$d_H(S_z, S_{z'}) \leq \Delta/s_0. \quad (9)$$

The differential maps $T_v S_z$ onto $T_{\Phi(v)} S_{z'}$, and

$$e^{-kC_A\Delta} \leq J_{\text{fib}}\Phi_{z,z'}(v) \leq e^{kC_A\Delta}. \quad (10)$$

Proof. With $A(v) = DH(v)^\dagger$, $DH(v)A(v) = I_r$, so every solution satisfies

$$\frac{d}{dt}H(v(t)) = \xi, \quad H(v(t)) = z + t\xi.$$

Convexity keeps the observation path in G , and Definition 5.1 keeps the solution in the compact regular tube. The boundary-submersion part of Assumption 5.2 gives the asserted neat level sets. ODE existence and uniqueness give the flow through time one; reversing ξ reverses the trajectory. Since $\|A\| \leq 1/s_0$, integration proves (6), and applying it in both directions proves (9).

Lemma 5.3 bounds the vector-field derivative by $C_A\Delta$. The variational equation for $Y(t) = D\Phi_t(v)$ and Gronwall's inequality give $\|Y(t)\| \leq e^{C_A\Delta t}$, which proves (7). Its integral form also gives

$$\|Y(1) - I_d\| \leq \int_0^1 C_A\Delta e^{C_A\Delta t} dt = e^{C_A\Delta} - 1,$$

proving (8). Differentiating $H(\Phi(v)) = H(v) + \xi$ carries $\ker DH(v)$ into $\ker DH(\Phi(v))$; invertibility gives equality. Apply the derivative bound to both flow directions. Every tangential singular value then lies between $e^{-C_A\Delta}$ and $e^{C_A\Delta}$; their product is (10). $\square$

For $H(x, y) = ax$, fiber displacement is exactly $|z - z'|/|a|$, so the $1/s_0$ dependence cannot be removed.

6 Conditional measures and their transport

Define

$$J_H(v) = \sqrt{\det(DH(v)DH(v)^\top)}, \quad Q(z) = \int_{S_z} \frac{\rho(v)}{J_H(v)} d\mathcal{H}^k(v).$$

Under Assumption 5.2, the coarea theorem gives a canonical fiber law

$$\nu_z(dv) = \frac{\rho(v)J_H(v)^{-1}}{Q(z)} d\mathcal{H}^k|_{S_z}(v), \quad (11)$$

which agrees with $\mathcal{L}(V \mid H(V) = z)$ for $H_\# \mathcal{L}(V)$ -almost every z . This formula is standard and is not claimed as a contribution [1].

Theorem 6.1 (Conditional-measure transport). *Let $\tilde{\nu}_{z'} = (\Phi_{z,z'})_\# \nu_z$ and set*

$$L_w = \frac{L_\rho}{\rho_{\min}} + \frac{rL_H}{s_0}, \quad C_0 = \frac{L_w}{s_0} + kC_A.$$

Under Assumption 5.2, for $\Delta = \|z' - z\|$,

$$\left\| \log \frac{d\nu_{z'}}{d\tilde{\nu}_{z'}} \right\|_{\infty} \leq 2C_0\Delta, \quad (12)$$

$$\|\nu_{z'} - \tilde{\nu}_{z'}\|_{\text{TV}} \leq \frac{1}{2}(e^{2C_0\Delta} - 1), \quad (13)$$

$$W_1(\nu_z, \nu_{z'}) \leq \frac{\Delta}{s_0} + \frac{D_{\mathcal{K}}}{2}(e^{2C_0\Delta} - 1). \quad (14)$$

For $p \geq 1$,

$$W_p(\nu_z, \nu_{z'}) \leq \frac{\Delta}{s_0} + D_{\mathcal{K}} \left[\frac{1}{2}(e^{2C_0\Delta} - 1) \right]^{1/p}.$$

No Wasserstein-2 Lipschitz conclusion is asserted from this estimate.

Proof. Put $w = \rho/J_H$. Matrix differentiation gives

$$D \log J_H(v)[h] = \text{tr}(DH(v)^{\dagger} D^2 H(v)[h]).$$

The rank bound for the trace and the density lower bound imply $\|\nabla \log J_H\| \leq rL_H/s_0$ and $\|\nabla \log \rho\| \leq L_{\rho}/\rho_{\min}$. Along the horizontal trajectory $v(t)$ from v to $\Phi(v)$, rather than along an ambient line segment,

$$|\log w(\Phi(v)) - \log w(v)| \leq \int_0^1 \|\nabla \log w(v(t))\| \|\dot{v}(t)\| dt \leq \frac{L_w}{s_0} \Delta.$$

No global Lipschitz conclusion on the possibly nonconvex or disconnected set $\mathcal{K}_G$ is used. Together with (10), this yields

$$e^{-C_0\Delta} \leq \frac{w(\Phi(v))J_{\text{fib}}\Phi(v)}{w(v)} \leq e^{C_0\Delta}.$$

Fiber change of variables gives

$$Q(z') = \int_{S_z} w(\Phi(v))J_{\text{fib}}\Phi(v) d\mathcal{H}^k(v),$$

so the same bounds hold for $Q(z')/Q(z)$. At $y = \Phi(v)$,

$$\frac{d\nu_{z'}}{d\tilde{\nu}_{z'}}(y) = \frac{w(y)J_{\text{fib}}\Phi(v)Q(z)}{w(v)Q(z')}.$$

This proves (12); integrating the density-ratio error proves (13).

Couple $v \sim \nu_z$ with $\Phi(v)$ to obtain $W_p(\nu_z, \tilde{\nu}_{z'}) \leq \Delta/s_0$. On a set of diameter $D_{\mathcal{K}}$, $W_p \leq D_{\mathcal{K}} \text{TV}^{1/p}$. The triangle inequality proves (14) and the general p bound. $\square$

The TV distance in (13) compares laws after putting them on the same transported fiber. Direct TV continuity between ν_z and $\nu_{z'}$ in ambient space is generally false because distinct fibers are disjoint.

Corollary 6.2 (Conditional data-law regularity). *For $\mu_z = F_{\#}\nu_z$,*

$$W_1(\mu_z, \mu_{z'}) \leq \frac{M_F}{s_0} \Delta + \frac{D_F}{2} (e^{2C_0\Delta} - 1).$$

A random-mask version follows by conditioning separately on each mask, provided that each conditional prior $V \mid M = m$ and map H_m satisfy Assumption 5.2, including its positive C^1 density hypothesis. The constants may depend on the mask.

Proof. Split through $F_{\#}\tilde{\nu}_{z'}$. The coupling $v \mapsto \Phi(v)$ gives

$$\begin{aligned} W_1(F_{\#}\nu_z, F_{\#}\tilde{\nu}_{z'}) &\leq \int_{S_z} \|F(\Phi(v)) - F(v)\| \nu_z(dv) \\ &\leq M_F \int_{S_z} \int_0^1 \|\dot{v}(t)\| dt \nu_z(dv) \leq \frac{M_F}{s_0} \Delta. \end{aligned}$$

Both $F_{\#}\tilde{\nu}_{z'}$ and $F_{\#}\nu_{z'}$ are supported in the set $F(\mathcal{K}_G)$ of diameter D_F . Hence their Wasserstein distance is at most D_F times their total-variation distance. Total variation contracts under pushforward, so (13) and the triangle inequality prove the claim. $\square$

Remark 6.3 (Why an ambient derivative bound is not a global Lipschitz bound). *On two disconnected components, a C^1 map may be constant with different values on the two components. Then $\sup \|DF\| = 0$ locally on the domain while the global Lipschitz constant is positive. This is why the transported-motion term above uses M_F only along actual flow trajectories, whereas the reweighted-mass term uses the image diameter D_F .*

7 Image manifolds and Gaussian smoothing

Assume $k = d - r \geq 1$ and suppose the intrinsic tangent noncollapse condition

$$\tau_0 = \inf_{v \in \mathcal{K}_G} \inf_{\substack{u \in \ker DH(v) \\ \|u\|=1}} \|DF(v)u\| > 0 \quad (15)$$

and $F|_{S_z}$ is injective for every $z \in G$.

Theorem 7.1 (Motion and curvature of conditional image manifolds). *Under Assumption 5.2, (15), and the stated injectivity of $F|_{S_z}$ for every $z \in G$, $\mathcal{M}_z = F(S_z)$ is an embedded C^2 manifold, possibly with boundary, of dimension k . The conjugated map*

$$\Psi_{z,z'} = F \circ \Phi_{z,z'} \circ (F|_{S_z})^{-1}$$

is a diffeomorphism $\mathcal{M}_z \rightarrow \mathcal{M}_{z'}$ and

$$\|\Psi_{z,z'} - \text{Id}\|_{\infty} \leq (M_F/s_0)\Delta, \quad (16)$$

$$\|D\Psi_{z,z'} - I\|_{T\mathcal{M}_z \rightarrow \mathbb{R}^D} \leq \frac{1}{\tau_0} \left[M_F(e^{C_A\Delta} - 1) + \frac{M_{F,2}\Delta}{s_0} e^{C_A\Delta} \right]. \quad (17)$$

Moreover,

$$\|\Pi_{\mathcal{M}_z}\|_{\text{op}} \leq \frac{M_{F,2} + M_F L_H / s_0}{\tau_0^2}.$$

Proof. Tangent noncollapse makes $F|_{S_z}$ an immersion. A continuous injective immersion from compact S_z to Euclidean space is an embedding. Conjugating the fiber diffeomorphism proves the first claim and the displacement bound. For $u \in T_v S_z$,

$$D\Psi(DF(v)u) - DF(v)u = [DF(\Phi(v))D\Phi(v) - DF(v)]u.$$

Along the horizontal trajectory from v to $\Phi(v)$,

$$DF(\Phi(v)) - DF(v) = \int_0^1 D^2F(v(t))[\dot{v}(t)] dt, \quad \|DF(\Phi(v)) - DF(v)\| \leq \frac{M_{F,2}}{s_0} \Delta.$$

Combining this bound with (7)–(8) and $\|(DF|_{T_v S_z})^{-1}\| \leq 1/\tau_0$ gives the derivative estimate without requiring the Euclidean chord from v to $\Phi(v)$ to remain in the regular region.

Differentiating H twice in a fiber chart gives

$$\Pi_{S_z}(u, w) = -DH(v)^\dagger D^2 H(v)[u, w],$$

hence $\|\Pi_{S_z}\| \leq L_H/s_0$. The second derivative of $F|_{S_z}$ is bounded by $M_{F,2} + M_F L_H/s_0$. Unit image tangents lift with norm at most $1/\tau_0$, and normal projection does not increase norm, proving the curvature bound. $\square$

Curvature does not imply a positive uniform reach: distant parameter points can have nearby images. No reach claim is made without global separation.

Proposition 7.2 (Gaussian-smoothed conditional density). *For $\alpha \in \mathbb{R}$, $\sigma > 0$, and the centered Gaussian density*

$$\varphi_\sigma(x) = (2\pi\sigma^2)^{-D/2} \exp\left(-\frac{\|x\|^2}{2\sigma^2}\right),$$

define

$$p_\sigma(x | z) = \int \varphi_\sigma(x - \alpha y) \mu_z(dy).$$

Then

$$\|p_\sigma(\cdot | z) - p_\sigma(\cdot | z')\|_1 \leq \sqrt{2/\pi} \frac{|\alpha|}{\sigma} W_1(\mu_z, \mu_{z'}), \quad (18)$$

$$\|\nabla p_\sigma(\cdot | z) - \nabla p_\sigma(\cdot | z')\|_1 \leq \frac{|\alpha| \sqrt{D+1}}{\sigma^2} W_1(\mu_z, \mu_{z'}). \quad (19)$$

Proof. For any coupling (Y, Y') , translate the Gaussian along the segment between αY and $\alpha Y'$. Fubini's theorem and rotational invariance give

$$\|\varphi_\sigma(\cdot - \alpha Y) - \varphi_\sigma(\cdot - \alpha Y')\|_1 \leq |\alpha| \|Y - Y'\| \sqrt{2/\pi}/\sigma.$$

For gradients, use the same argument with $D^2 \varphi_\sigma(x)u$. If $G \sim \mathcal{N}(0, I_D)$, Cauchy–Schwarz yields

$$\int \|D^2 \varphi_\sigma(x)u\| dx \leq \sigma^{-2} \left[\mathbb{E} \|(GG^\top - I)u\|^2 \right]^{1/2} = \sqrt{D+1}/\sigma^2.$$

Average and minimize over couplings. $\square$

These estimates do not prove score regularity for $\nabla p_\sigma/p_\sigma$; Gaussian tails create a denominator problem.

8 Conditional diffusion entropy corollary

Corollary 8.1 (Version-pinned pointwise DDPM entropy input). *Use the random-mask setting of Corollary 3.3. On a Borel good set of regular observations, put $k(m, z) = 2n_v - r(m, z) \geq 1$. For almost every fixed good (m, z) , assume the remaining conventions and hypotheses of Theorem 2 of Liang et al. [10, arXiv version 2], and let (X_t) and (Y_t) denote, respectively, the forward and original-DDPM chains defined there. In the notation of that theorem, assume the conditional score errors*

$$\varepsilon_{\text{score},t}^2(m, z) = \mathbb{E}[\|s_t(X_t; m, z) - s_t^*(X_t; m, z)\|^2 | Z = (m, z)]$$

satisfy its unweighted condition

$$\frac{1}{T} \sum_{t=1}^T \varepsilon_{\text{score},t}^2(m, z) \leq \varepsilon_{\text{score}}^2(m, z).$$

Then, for all sufficiently large T (with the threshold allowed to depend on the fixed observation),

$$\text{TV}(p_{X_1|Z=(m,z)}, p_{Y_1|Z=(m,z)}) \lesssim_{m,z} \frac{k(m, z) \log^3 T}{T} + \varepsilon_{\text{score}}(m, z) \sqrt{\log T}.$$

Proof. Fix (m, z) outside the exceptional null set, abbreviate its exponent by $k \geq 1$, and choose the single scale $\varepsilon_0 = T^{-c_{\varepsilon_0}}$ required by Assumption 1 of Liang–Huang–Chen v2. Corollary 3.3 gives

$$\log \mathcal{N}_{\varepsilon_0} \leq C''_{m,z} + k c_{\varepsilon_0} \log T \leq \tilde{C}_{m,z} k \log T$$

for $T \geq 2$, after absorbing the fixed additive constant. This is their single-scale intrinsic-dimension condition. The conditional support is compact, so its fixed radius is at most T^{c_R} for all sufficiently large T , as required by their Assumption 2. The displayed score condition is their Assumption 3. Applying their Theorem 2 to this fixed conditional target law gives the stated pointwise bound. $\square$

The source EIT paper invokes the same single-scale result in its Theorem 4.15. We pin arXiv version 2 because its hypotheses are the ones verified in the preceding argument; adapting the corollary to subsequent formulations is left for future work. The result remains external and pointwise conditional: neither its score-error hypothesis nor constants uniform in (m, z) are established here. When $r = 2n_v$, the fiber is finite, but we do not substitute a zero-dimensional parameter into an external theorem formulated for positive intrinsic dimension.

9 Discretized EIT model and numerical methodology

The numerical examples use a unit-disk finite-element model with piecewise linear basis functions. A conductivity equals one outside a convex quadrilateral and four inside it. The four counterclockwise vertices are indexed by a fixed template convention, giving a parameter in $\mathbb{R}^8$. The admissible generator enforces positive margins for side length, distance from the disk boundary, and interior angles. This finite-dimensional chart is the object differentiated numerically; the experiments do not establish the regularity of the corresponding continuum shape map [2].

The principal mesh has 145 vertices, 256 triangles, and 32 boundary nodes. The discrete DtN matrix is symmetric, and its 528 upper-triangular entries form the complete data vector. Entrywise background normalization fixes the observation metric. A mask selects rows of the resulting Jacobian; all reported singular values therefore refer to the Euclidean vertex metric and the declared normalized or noise-whitened observation metric. The synthetic population was generated with seed 20260901 and separated into 2,000 training, 200 validation, and 400 test polygons. The full population data are not redistributed in this repository; the retained numerical files are compact, non-personal diagnostic outputs.

Central differences use the nondimensionalized parameter step $h = 0.002$ and a relative singular-value threshold of 10^{-3} unless stated otherwise. A step-size sweep from 10^{-2} to 10^{-6} checks the stability of this choice. Candidate masks are selected without validation or test polygons. The fiber experiments use two constructions: predictor–corrector continuation initialized in a numerical nullspace, and constrained correction from isotropic perturbations in the full $\mathbb{R}^8$ parameter space. The latter serves as an independent control on the dimension supplied to the first method. Local PCA

uses a 10^{-2} relative eigenvalue threshold. Greedy covering estimates are calibrated on Euclidean balls of known dimension at identical sample sizes.

The mesh comparison maps the same physical boundary-coordinate pairs to three refinements. A separate complete-electrode-model (CEM) derivative diagnostic uses reference software and mesh material distributed with the KTC 2023 v3 dataset [13]; only aggregate derivative diagnostics are retained here. No KTC code, mesh, or measurement data are redistributed. The source dataset and its license are identified in the repository’s third-party notice.

10 Numerical results

The experiments isolate predictions of the preceding analysis rather than testing a conductivity-reconstruction endpoint. They provide finite- dimensional numerical evidence and do not replace any regularity assumption or proof.

10.1 Independent fiber recovery and rank along sampled fibers

The first continuation method used predictors in the numerical nullspace of a base Jacobian. To remove construction-induced circularity, a second method started isotropically in the full $\mathbb{R}^8$ parameter space and corrected the observation residual without nullspace initialization. Both methods recovered the dimensions shown in Table 1.

Table 1: Independent local fiber recovery. PCA uses a 10^{-2} relative eigenvalue threshold.

thresholded rank	predicted $8 - r$	nullspace start	full- $\mathbb{R}^8$ start
2	6	6	6
4	4	4	4
6	2	2	2

The full-space experiment was repeated on ten held-out polygons stratified by admissibility margin. All 30 recovered clouds matched the predicted PCA dimension. Centered-difference Jacobians were recomputed at all accepted points: 1,138 of 1,138 retained the expected thresholded full row rank. This checks numerical rank separation along the cloud; it does not certify exact differential rank. For the endpoint $r = d = 8$, 96 nearby full-space starts returned to one local solution, with maximum distance 1.1609×10^{-7} from the base point.

10.2 Full spectrum, criticality, covering bias, and mesh sensitivity

Across six right-singular directions, four residual radii, and both signs, the 48 quantities $\sigma_i W_i / \delta$ had mean 0.999961 and range $[0.998971, 1.000797]$. On 48 separate rank-six linear ellipsoids, the full-spectrum logarithmic cover predictor had mean usable Spearman correlation 0.990 with greedy cover count, versus 0.854 for a surrogate that repeats only $\sigma_{\min}$.

A same-rank near-critical mask had $\sigma_6 = 3.63 \times 10^{-4}$, versus 0.263 for the stable mask, and amplified mean linearized parameter error by a factor of 402.7. This is a near-critical stability control, not evidence of exact rank change or fiber branching.

Finite-cloud covering slopes are downward biased. Applying the identical estimator to 193 points from known Euclidean balls embedded in $\mathbb{R}^8$ gives Table 2; therefore raw EIT slopes are not treated as consistent continuum-dimension estimates.

Table 2: Greedy-cover calibration at 193 points; mean $\pm$ standard deviation over four independent clouds.

true dimension	estimated slope	downward bias
2	1.507 ± 0.037	24.6%
4	2.629 ± 0.093	34.3%
6	3.647 ± 0.242	39.2%

Mapping the same six physical boundary-coordinate pairs to three disk-mesh refinements preserved thresholded rank six. The normalized weakest-direction widths were 0.9986, 0.9996, and 0.9988; the normalized spectrum changed by 0.0819 between the primary and finest mesh. This is a discretization sensitivity check, not a continuum FEM convergence theorem.

11 Supporting identities and interface checks

The following results are standard supporting tools. They are included for completeness and are not presented as contributions.

Proposition 11.1 (Nonexpansion of compatible affine projection). *Let $\mathcal{C}$ be a nonempty closed affine subset of a finite-dimensional Hilbert space and let $\Pi_{\mathcal{C}}$ be its orthogonal projection. If the truth $X^* \in \mathcal{C}$, then every estimate $\hat{X}$ satisfies*

$$\|\Pi_{\mathcal{C}}\hat{X} - X^*\| \leq \|\hat{X} - X^*\|.$$

Proof. Orthogonal projection onto a nonempty closed convex set obeys the variational inequality $\langle \hat{X} - \Pi_{\mathcal{C}}\hat{X}, X^* - \Pi_{\mathcal{C}}\hat{X} \rangle \leq 0$. Expanding the squared distance from $\hat{X}$ to X^* gives the result. $\square$

In 20,000 compatible random cases, the numerical inequality held to the stated tolerance. In a noisy incompatible control, hard projection either failed to improve the distance or violated compatibility in all 20,000 cases. The control illustrates why compatibility of the truth is required.

Proposition 11.2 (Linear-Gaussian missing-response MMSE). *Let $\eta \sim \mathcal{N}(0, I_\ell)$, $y = \mu + B\eta$, and observe*

$$\tilde{y}_M = \mu_M + B_M\eta + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \Gamma_M),$$

independently, with Γ_M positive definite. Put

$$P_M = (I_\ell + B_M^\top \Gamma_M^{-1} B_M)^{-1}.$$

Then the Bayes risk for squared-error prediction of the missing response is

$$R_M = \mathbb{E}\|y_{\bar{M}} - \mathbb{E}[y_{\bar{M}} \mid \tilde{y}_M]\|^2 = \text{tr}(B_{\bar{M}} P_M B_{\bar{M}}^\top).$$

Proof. Gaussian conditioning, or completion of the square, gives posterior covariance $\text{Cov}(\eta \mid \tilde{y}_M) = P_M$. The posterior covariance of $y_{\bar{M}}$ is therefore $B_{\bar{M}} P_M B_{\bar{M}}^\top$. The conditional mean is the squared-error Bayes estimator, and its mean squared error is the trace of this covariance. $\square$

With 200,000 samples, the theoretical risk was 0.1307554 and the empirical risk was 0.1312530, a relative difference of 0.381% and a Monte Carlo z -score of 1.39.

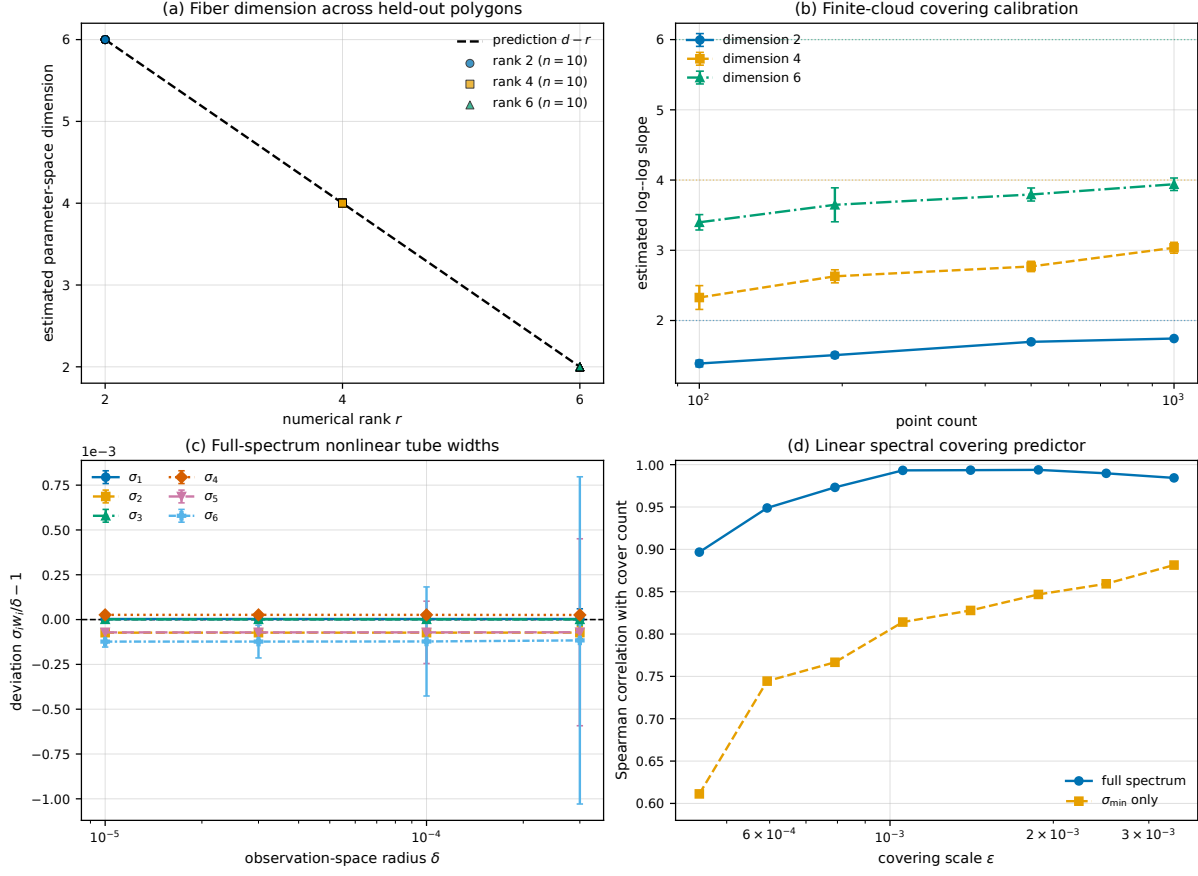


Figure 1: Numerical tests of the local geometric mechanisms. Top left: independent full-dimensional fiber recovery for ten held-out polygons; top right: finite-cloud covering calibration; bottom left: deviations of the normalized singular-direction widths $\sigma_i W_i / \delta$ from one; bottom right: full-spectrum and weakest-singular-value covering predictors.

Proposition 11.3 (Smooth finite-dimensional CEM voltage chart). *Let a bounded Lipschitz domain have fixed disjoint electrodes of positive surface measure and positive contact impedances. Suppose $\theta \mapsto \gamma_\theta$ is C^1 from an open subset of $\mathbb{R}^p$ to L^∞ , with conductivity uniformly bounded above and away from zero on compact parameter subsets. For fixed gauge-normalized current patterns, the finite vector of complete-electrode-model voltages is C^1 in θ .*

Proof. On the gauge-fixed space $\mathcal{H} = H^1(\Omega) \oplus \mathbb{R}_0^L$, the standard CEM bilinear form is

$$a_\theta((u, U), (v, V)) = \int_{\Omega} \gamma_\theta \nabla u \cdot \nabla v \, dx + \sum_{j=1}^L z_j^{-1} \int_{E_j} (u - U_j)(v - V_j) \, ds.$$

The conductivity and impedance bounds, trace theorem, and gauge condition give uniform boundness and coercivity on compact parameter subsets [14]. Its operator $A(\theta) : \mathcal{H} \rightarrow \mathcal{H}^*$ is C^1 . The identity $D(A^{-1})[h] = -A^{-1}(DA[h])A^{-1}$ makes the weak solution C^1 , and finite voltage readout is bounded linear. $\square$

The implemented four-dimensional CEM chart had a finite-difference plateau change below 1.16×10^{-5} , relative discrepancy 0.00124 from the reference-solver Jacobian at the reference step,

and directional Taylor residual slope 1.976. This checks the software interface in a smooth coefficient chart; it does not establish shape differentiability for moving discontinuous polygons.

The anisotropic whitening control used 500 masks and 400 trials per mask. The Spearman correlation between inverse singular value and observed amplification was 0.413 in raw coordinates and 0.988 after whitening. In a separate selection study, the unweighted design error was 3.93 times the whitened design error. These experiments support the whitening prescription in Corollary 4.2; they do not make whitening a new theorem.

12 Analytic sanity checks

The numerical package tests four transparent cases. For $H(x, y) = x$, the transport is $(z, y) \mapsto (z', y)$ and the motion bound is exact with $s_0 = 1$. For $H(x, y) = ax$, displacement is $|z - z'|/|a|$. For $H(x, y) = x + y^2$,

$$DH(x, y)^\dagger = \frac{(1, 2y)^\top}{1 + 4y^2},$$

and the horizontal ODE preserves $H(v(t)) - t\xi$ exactly. Finally, $H(x, y) = x^2$ has $s_0 = 2|x|$; the lift norm diverges at $x = 0$, the rank collapses, and no uniform Lipschitz transport crosses the critical value.

13 Discussion and limitations

The mathematical results form the following chain. Exact conditioning localizes the population law to a compact subset of a $(d - r)$ -dimensional regular level set; noisy finite-resolution complexity resolves the complete whitened singular spectrum; quantitative horizontal flow transports regular fibers; and coarea weighting turns this into conditional-law Wasserstein-1 regularity. At the single scale required by the explicitly pinned arXiv-v2 external DDPM theorem, the entropy refinement supplies its covering input for almost every fixed regular observation, conditional on that theorem's other hypotheses. The EIT experiments are consistent with the predicted local mechanisms.

The result is deliberately narrower than a complete conditional-diffusion theory. The project has not established whole-prior EIT flow completeness, uniform reach, conditional-score regularity, or finite-sample score estimation. Those items are excluded from the claim set and recorded only in the repository future-work document.

References

- [1] Fatima-Zahrae Akhyar, Wei Zhang, Gabriel Stoltz, and Christof Schütte. Generative modeling of conditional probability distributions on the level-sets of collective variables. *arXiv preprint arXiv:2512.17374*, 2025.
- [2] Elena Beretta, Elisa Francini, and Sergio Vessella. Differentiability of the dirichlet to neumann map under movements of polygonal inclusions with an application to shape optimization. *SIAM Journal on Mathematical Analysis*, 49(2):756–776, 2017. doi: 10.1137/16M1082160.
- [3] Elena Beretta, Elisa Francini, and Sergio Vessella. Lipschitz stable determination of polygonal conductivity inclusions in a two-dimensional layered medium from the dirichlet-to-neumann map. *SIAM Journal on Mathematical Analysis*, 53(4):4303–4327, 2021. doi: 10.1137/20M1369609.

- [4] Tan Bui-Thanh, Qin Li, and Leonardo Zepeda-Núñez. Bridging and improving theoretical and computational electrical impedance tomography via data completion. *SIAM Journal on Scientific Computing*, 44(3):B668–B693, 2022. doi: 10.1137/21M141703X.
- [5] Ke Chen, Haizhao Yang, and Chugang Yi. Data completion for electrical impedance tomography by conditional diffusion models. *arXiv preprint arXiv:2602.07813v1*, 2026. URL <https://arxiv.org/abs/2602.07813v1>. Version 1, submitted 8 February 2026.
- [6] Bastian Harrach. Uniqueness and lipschitz stability in electrical impedance tomography with finitely many electrodes. *Inverse Problems*, 35(2):024005, 2019. doi: 10.1088/1361-6420/aaf6fc.
- [7] N. Hyvönen, A. Jääskeläinen, R. Maity, and A. Vavilov. Bayesian experimental design for head imaging by electrical impedance tomography. *arXiv preprint arXiv:2312.10383*, 2024.
- [8] Ahmad Karimi, Leila Taghizadeh, and Clemens Heitzinger. Optimal bayesian experimental design for electrical impedance tomography in medical imaging. *Computer Methods in Applied Mechanics and Engineering*, 373:113489, 2021. doi: 10.1016/j.cma.2020.113489.
- [9] John M. Lee. *Introduction to Smooth Manifolds*. Springer, 2 edition, 2013.
- [10] Jiadong Liang, Zhihan Huang, and Yuxin Chen. Low-dimensional adaptation of diffusion models: Convergence in total variation. *arXiv preprint arXiv:2501.12982v2 [stat.ML]*, 2025. URL <https://arxiv.org/abs/2501.12982v2>. Version 2, revised 16 June 2025; Assumptions 1–3 and Theorem 2.
- [11] Sven Nordebo, Mats Gustafsson, Börje Nilsson, Therese Sjöden, and Francesco Soldovieri. Fisher information analysis in electrical impedance tomography. *Journal of Geophysics and Engineering*, 10(6):064008, 2013. doi: 10.1088/1742-2132/10/6/064008.
- [12] Claire C. Onsager, Chulin Wang, Charles Costakis, Can Aygen, Lauren Lang, Suzan van der Lee, and Matthew A. Grayson. Sensitivity volume as figure-of-merit for maximizing data importance in electrical impedance tomography. *Physiological Measurement*, 2024. doi: 10.1088/1361-6579/ad3458. Preprint arXiv:2111.01397.
- [13] Mikko Räsänen, Petri Kuusela, Jyrki Jauhiainen, Muhammad Arif, Kenneth Scheel, Tuomo Savolainen, and Aku Seppänen. Kuopio tomography challenge 2023 open electrical impedance tomographic dataset (KTC 2023), 2024. URL <https://doi.org/10.5281/zenodo.10986692>. Dataset and reference software, licensed under CC BY 4.0.
- [14] Erkki Somersalo, Margaret Cheney, and David Isaacson. Existence and uniqueness for electrode models for electric current computed tomography. *SIAM Journal on Applied Mathematics*, 52(4):1023–1040, 1992. doi: 10.1137/0152060.
- [15] Roman Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge University Press, 2018.